

Exploring the Potential of Psilocybin: A Review of its Effects on Depression

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Abstract

Depression is a global burden, affecting millions of individuals worldwide. Psilocybin, a psychedelic substance, is being explored as a potential drug for alleviating symptoms associated with depressive disorders. This review aims to evaluate the viability of psilocybin therapy for the treatment of depression, with the hypothesis that psilocybin has a negative effect on depressive symptoms. To support this hypothesis, empirical evidence from four different studies is investigated. These studies encompass both healthy individuals and those facing life-threatening illnesses as well as individuals who are diagnosed with a Major Depressive Disorder. All the findings from these studies align with the hypothesis, revealing that psilocybin has a negative effect on depressive symptoms. It can therefore be concluded that psilocybin holds promise as a potential treatment for depression, offering a means to improve the well-being of individuals affected by this condition.

Exploring the Potential of Psilocybin: A Review of its Effects on Depression

The increasing prevalence of depression presents a pressing challenge in contemporary society. Research has attempted to unravel the intricate web of factors, including genetics, environment, and psychology, that contribute to its aetiology. While the exact causation is unknown, depression does appear to be a universal phenomenon that affects over 280 million people worldwide (World Health Organization, 2021) and was the global leading cause of burden in 2017 (James et al., 2018). The prevalence of depression varies greatly per country, with higher rates observed in predominantly Western and affluent nations (Kessler & Bromet, 2013). This defies the conventional belief that depression is solely a consequence of adverse life conditions. There is therefore a need for an alternative hypothesis to account for this regional disparity. One such hypothesis is that the individualization and secularization observed in Western societies (Cohen et al., 2016) play an important role in the higher rates of depression (Hawkley & Capitanio, 2015). Social institutions, such as religious affiliations and local communities, can be viewed as potent protective factors against social isolation and, consequently, mental health issues (Ronneberg et al., 2014; Adabla & Nabors, 2022). These institutions have long served as pillars of support by providing meaningful connections and fostering the development of shared values, norms, and beliefs. It was shown already in the 20th century, most notably by Sherif (1937) and Asch (1955), that people rely on and/or utilise others to make sense of the world and continuously refine their values. With the erosion of these previously described institutions, an essential part of the human meaning-making machinery is lost. Without a functioning meaning-making process, individuals may feel a profound disconnection, potentially contributing to the development of depressive symptoms (Marco et al., 2020). One product of this process is the subjective sense of “realness”. Often when an individual refers to something as “real”, it is not meant as a verifiable mathematical-like truth. Instead, it is used

to describe a connection between a subjective experience and the observed patterns in the world (Vervaeke et al., 2017). A sense of realness instead should be classified as a psychological experience and not as a metaphysical truth. These are important mechanisms for healthy individuals, and when disturbed, can greatly undermine their functioning.

A previous study by Hendricks et al. in 2015 has shown that psychedelics are associated with the reduction of psychological distress and suicidality in the United States adult population. High levels of psychological distress and suicidal thoughts can be indicators of impaired mental health and may reflect symptoms of common mental disorder classifications in the DSM-5-TR, like depressive and anxiety disorders (American Psychiatric Association, 2022). This study might therefore be indicative of the anti-depressant effects of psychedelics. Psychedelics are a class of psychoactive substances that alter an individual's perception, cognition, and mood. They produce profound changes in consciousness, often characterized by vivid sensory experiences, alterations in thinking patterns, and a sense of expanded awareness (Nichols & Nichols, 2021; Yaden et al., 2021). Psilocybin is a naturally occurring psychedelic that is found in over 200 species of mushrooms (Guzmán et al., 1998) as well as in fungal sclerotia commonly called “magic truffles” (Pellegrini et al., 2012). A study by Grob in 2010 investigated the effects of psilocybin on treating anxiety in advanced-stage cancer patients. The results showed that the administration of a moderate dose of psilocybin had a negative effect on anxiety. As mentioned before, social connections and meaningful relationships are protective factors against mental health issues. Similarly, with psilocybin, individuals often experience a profound sense of connection and unity with others, nature, and reality itself (Studerus et al., 2010). This increased feeling of connectedness and meaning during the experience could potentially have a lasting impact which in turn might counteract some of the lost protective factors.

These studies show that psychedelics have the potential for reducing psychological distress and anxiety. They also hint at the possibility of using psilocybin as an anti-depressant, this however is not evident from these studies and will be the topic of this literature review. The Entropic Brain Theory (EBT; Carhart-Harris et al., 2014) could serve as an explanatory theory for psilocybin's anti-depressant potential. EBT explains the psychedelic-induced altered state of consciousness by describing the amount of entropy in the brain. It has been observed through neuroimaging data that the amount of entropy in the brain increases when psilocybin is administered (Carhart-Harris et al., 2014). This elevation of entropy is manifested by an increase of variance in the neural activity of high-level association networks. These high-level association networks include the Default Mode Network (DMN) which is associated with introspective thought and self-reference (Andrews-Hanna et al., 2010). More recently, the DMN is proposed to function as a general "sense-making" network (Yeshurun et al., 2021). Depression has been observed to manifest as hyperactivity and dysconnectivity within this network, as indicated by studies conducted by Yan et al. (2019) and Berman et al. (2010). In other words, depression might undermine the "sense-making" network and would as a consequence, disrupt the subjective sense of connection between oneself and the external world. Psilocybin could afford the transition out of this state by inducing a flexible neurological state through its ability to increase variance in neural activity (Carhart-Harris et al., 2014). This increased neural flexibility could in turn facilitate the normalization of the DMN, which would simultaneously normalize the meaning-making process and re-establish the subjective sense of connectedness. By means of this mechanism, the hypothesis is that psilocybin has a negative effect on depressive symptoms. The viability of this hypothesis will be explored in this literature review.

This review is structured to provide a comprehensive account of the effects of psilocybin on depression. The dosage labels of psilocybin are generally the same across the

mentioned studies, but some differ slightly in amount. The following classifications are used: placebo-like dose, moderate-dose, and high-dose. All studies determine the dosage by the weight of the participant. The placebo-like dose is 1mg/70kg (also referred to as “microdose”), the moderate-dose is 20mg/70kg and the high-dose is 30mg/70kg. The first study will give a partial neurological account of the changes that occur during a psilocybin experience. Understanding the neural systems involved in depression and the role of medication is crucial for developing effective forms of treatment (Pankevich et al., 2014). One brain region that has been extensively studied in relation to depression is the amygdala, which has been implicated in depression and other internalising disorders. Research suggests that amygdala hyperreactivity may contribute to an increased risk for depression, although a recent study has only shown negligible or small effects (Grogans et al., 2022). While the findings for a relationship between amygdala hyperreactivity and depression are mixed, it is still proposed to play a role in depression as Major Depressive Disorder (MDD) is associated with larger right-sided medial nuclei amygdala volumes (Roddy et al., 2021). For this reason, the first study will examine the effects of psilocybin on amygdala reactivity in healthy participants. Subsequently, a study focused on cancer patients with life-threatening diagnoses explored the effects of psilocybin in reducing depressed mood and anxiety. The third study examines the effectiveness of psilocybin therapy specifically in patients diagnosed with MDD. The final study also investigated the effects of psilocybin-assisted treatment for people diagnosed with MDD but with an extended follow-up to see if the effects would sustain for a longer period. These studies will provide an overview of both self-reported and objectively measured effects of psilocybin on depression, covering healthy individuals and most importantly, people who are diagnosed with MDD.

Empirical Evidence

In a study from 2015 conducted by Kraehenmann et al., the effects of psilocybin were tested on amygdala reactivity. The effects of psilocybin were tested on 25 healthy participants between 20 and 27 years old. The participants were randomly divided into two conditions. The first condition (psilocybin condition) received a dose of psilocybin (11.2mg/70kg), while the second condition was administered a placebo (placebo condition). The selected dosage of psilocybin can be classified as a low- to moderate-dose; however, the study did not provide any rationale or label for the chosen dosage. Both conditions underwent two fMRI imaging sessions that were scheduled at least 14 days apart. During the two sessions, the participants would perform a slightly modified version of the amygdala reactivity task (Hariri et al., 2002) and a simple motor task to remove possible confounds. The amygdala reactivity during the session was assessed by measuring BOLD signals from both amygdalae. Participants' mood state was assessed using the Positive and Negative Affect Schedule (PANAS; Watson et al., 1988) and the state portion of the State-Trait Anxiety Inventory (STAI; Spielberger, 1983) before and 210 minutes after each administration. PANAS is a self-reported measure that evaluates positive (PANAS-P scale) and negative (PANAS-N scale) affect. Participants rate their experience of specific emotions by answering 20 items that are scored using a 5-point Likert scale. The STAI is a self-reported measure with 20 items that assess anxiety with the STAI-State (STAI-S) scale for temporary/state anxiety and STAI-Trait (STAI-T) scale for enduring anxiety. Participants rate their experience of emotions on a 4-point scale, from "Almost Never" to "Almost always". Results showed that the participants in the psilocybin condition scored higher on the PANAS-P scale than those in the placebo condition while the PANAS-N and STAI-S scores remained the same for both conditions. The measured BOLD signals from the amygdalae showed that participants in the psilocybin condition had reduced activation of the right amygdala for both negative and neutral pictures in comparison to the

placebo condition. The study concludes that psilocybin has a negative effect on amygdala reactivity while increasing positive affect for healthy participants.

This study examined the effects of psilocybin on amygdala reactivity and concluded that the observed effects may be indicative of antidepressant properties of psilocybin. It offers insight into a partial explanation of these properties from a neurological perspective. The study also hypothesises that possible anti-depressive effects can be explained by means of normalizing the amygdala's abnormal activity that is found in MDD. This phenomenon can be more comprehensively understood with the Entropic Brain Theory. In line with the proposed normalization of the DMN through enhanced neurological flexibility, it can be hypothesized that psilocybin facilitates the normalization of various brain regions, encompassing both the DMN and the amygdala. It is important to note that the effects of psilocybin were tested on healthy participants without a depressive disorder. Therefore, the results in this study regarding amygdala reactivity might not be generalisable to people who are diagnosed with a depressive disorder. The next study will focus on the effects of psilocybin on depressive symptoms for people with life-threatening cancer. Depression has strong comorbidity with people who are diagnosed with cancer (Smith, 2015) and would therefore be a more relevant group.

A study from 2016 conducted by Griffiths et al. included 51 cancer patients with life-threatening diagnoses who also had symptoms of depression and/or anxiety. Only a small subset of these participants, namely 18 out of 51, was diagnosed with MDD. The participants were randomly assigned to two conditions: a low-dose condition and a high-dose condition. The low-dose condition received a placebo-like dose of psilocybin (1mg/70kg, n=38; 3mg/70kg, n=12) in the first session and a high-dose (22mg/70kg, n=49; 30mg/70kg, n=1) in the second session, while the high-dose condition received the dosage in reverse. It should be noted that the dosage changed during the course of the experiment. Because this was one of

the earlier studies regarding this topic, little was known about the dosage. Due to an incident and concern, both the placebo-like dose and high-dose were lowered in amount. During both sessions, the participants completed the Beck Depression Inventory (BDI; Beck et al., 1961) and the Hospital Anxiety and Depression Scale (HADS; Zigmond & Snaith, 1983) self-report questionnaires which both aim to assess the amount and severity of depressive symptoms. The BDI consists of 21 items while the HADS includes 14 items. Additionally, a trained clinician conducted a semi-structured interview that contains 17 items from the GRID Hamilton Rating Scale (GRID-HAMD-17; ISCDD, 2003) to assess the participant's depressive symptoms. The BDI, HADS and GRID-HAMD-17 each use a 4-point scale per item with the total sum of scores indicating the severity of depressive symptoms. The results showed that a high dose of psilocybin produced significantly decreased scores in both the clinician-rated GRID-HAMD-17 and BDI and HADS questionnaires in comparison to the placebo-like dose. These effects were sustained at the 6-month follow-up, showing clinically significant decreases in depressed mood and anxiety. The study concluded that psilocybin has a negative effect on depressive symptoms in people with a life-threatening illness.

This study investigated the effects of psilocybin on depressive symptoms and showed that psilocybin can alleviate these symptoms in people with a life-threatening illness. Given that thoughts of mortality are more likely to occur in individuals confronting a life-threatening illness, it can be hypothesized that the negative perception of death in Western culture (Aries, 1974), contributes to the development of depressive and anxiety symptoms in this context. The induced cognitive flexibility by psilocybin as proposed by the Entropic Brain Theory, may facilitate the emergence of new perspectives, and enables individuals to re-evaluate their attitudes towards death in a positive light. It is important to note that because of the specific context of a potentially fatal illness, psilocybin might only help alleviate death

anxiety. Therefore, the next study will investigate the effects of psilocybin on depression in participants who are diagnosed with MDD and do not have a life-threatening illness.

Davis et al. conducted a study in 2021 which investigated the effectiveness of psilocybin therapy in patients diagnosed with MDD. The study enrolled adults aged 21 to 75 years with MDD, who were not already using antidepressant medications and do not have a history of hospitalisation due to other disorders. Participants were randomized into an immediate treatment condition or a delayed treatment condition. The intervention consisted of a moderate- and high-dosed psilocybin session that both included supportive psychotherapy. The immediate treatment condition started therapy directly, while the delayed treatment condition received treatment after an 8-week delay. The primary outcome measure was depression severity, assessed by a clinician using the GRID-HAMD-17 (ISCDD, 2003) in a semi-structured interview. The interview took place at baseline and weeks 5 and 8 for the delayed treatment condition while the immediate treatment condition was assessed in weeks 1 and 4. Secondary outcomes were measured using the Quick Inventory of Depressive Symptomatology-Self Rated (QIDS-SR; Rush et al., 2003) and Beck Depression Inventory II (BDI-II; Beck et al., 1996) questionnaires. The QIDS-SR includes 16 items, and the BDI-II has 21 items to assess the severity of depressive symptoms. Each item is scored using a 4-point scale and the sum of these scores indicates the severity of depression. Results showed that the participants in the immediate treatment condition had significantly lower GRID-HAMD-17, QIDS-SR, and BDI-II scores at weeks 1 and 4 compared to the participants in the delayed treatment condition. The study concluded that psilocybin in a therapeutic setting has a negative effect on depressive symptoms in people with MDD.

This study investigated the effects of psilocybin on depressive symptoms and showed that psilocybin can alleviate these symptoms in people with MDD. By testing the effectiveness on a diverse group of participants with MDD who do not have any other

disorders, the study has potentially high external validity. With results showing a reduction in symptoms for these participants, it provides further evidence for the effectiveness of psilocybin therapy in treating MDD. However, this study had a relatively short follow-up of 4 weeks. It is very common (>50% of the patients) to have periods of remission and relapse when diagnosed with MDD (Zaremba et al., 2018). One study suggests that relapse of MDD is largely due to genetic factors (Burcusa & Iacono, 2007). It is unsure whether the anti-depressive effects of psilocybin can overcome this biological predisposition. The proposed normalization of neural activity as explained by the Entropic Brain Theory might only be temporary in nature. In light of this, further delays in follow-ups are necessary. Therefore, the following study will investigate the effects of psilocybin MDD treatment with a follow-up of 12 months.

A study from 2022 by Gukasyan et al. investigated the long-term efficacy of psilocybin-assisted treatment for participants who were diagnosed with MDD. The study consisted of 27 participants who were randomly divided into an immediate or delayed treatment condition. In line with the study by Davis et al. from 2021, the participants received both a moderate and high weight-adjusted dose of psilocybin along with psychotherapy that was scheduled two weeks apart. The delayed treatment condition received treatment after 8 weeks while the immediate condition received treatment immediately. The effects were evaluated over a year with follow-ups at 1, 3, 6 and 12 months using the GRID-HAMD-17 (ISCDD, 2003) semi-structured interview by a clinician as well as the QIDS-SR (Rush et al., 2003) and BDI-II (Beck et al., 1996) questionnaires. The findings demonstrated significant reductions in scores for GRID-HAMD-17, QIDS-SR and BDI-II for both conditions at all follow-up points when compared to their baseline. The study concluded that psilocybin in a therapeutic setting has a negative effect on depressive symptoms for people diagnosed with MDD.

Conclusions and Discussion

The studies mentioned in this literature review all revealed that psilocybin has a negative effect on depressive symptoms for individuals with and without MDD. The first study indicated psilocybin's potential anti-depressive effects by examining amygdalae reactivity after administration. The study concluded that psilocybin has a negative effect on amygdala reactivity and hypothesized that psilocybin might be able to normalize hyperactivity in this region for individuals with MDD. The second study revealed psilocybin's effectiveness in the alleviation of anxiety and depressive symptoms in people with a life-threatening illness. The third and fourth studies demonstrated that psilocybin, when combined with psychotherapy, is effective in treating people with MDD. The final study complemented the third study by measuring the effects of psilocybin with extended follow-ups and showed that effects were sustained after 12 months. Conclusively, all findings align with the hypothesis that psilocybin has a negative effect on depressive symptoms. While the investigation of the underlying mechanisms of psilocybin is out of scope for this review, the Entropic Brain Theory would provide a theoretical explanation for the effects of psilocybin that are described in these studies. By enhancing neural flexibility, psilocybin might allow brain activity to normalize in regions such as the Default Mode Network and the amygdala. As a result, the normalization of the Default Mode Network might correct the process of meaning-making and restore the subjective sense of connectedness that is frequently diminished among individuals experiencing depression.

Although the studies in this review aimed to assess the effectiveness of psilocybin for therapeutic purposes, where psilocybin is described as the main factor, it is important to note that the administration of psilocybin alone does not necessarily entail therapeutic benefits. Instead, it can be hypothesized, that the psilocybin-induced flexible neurological state has the potential to have both adaptive as well as maladaptive outcomes depending on the

experience. It is found that the so-called “set and setting” is an important variable in determining this outcome (Hartogsohn, 2017). Another factor is supportive psychotherapy, which mainly facilitates “psychedelic integration”, which tries to properly integrate the experience in the context of one’s life (Bathje et al., 2022). Future studies could further investigate the optimal ways of facilitating and integrating these experiences.

The demonstrated negative relationship between psilocybin and depressive symptoms raises new questions about depressive disorders and traditional treatment approaches. Specifically, what are the precise abnormalities in a depressed brain, and how can these abnormalities be explained in light of the increased prevalence of depressive disorders? Secondly, is there a true link between depressive symptoms and these underlying neurological abnormalities? And if psilocybin were to normalize these abnormalities through induced neural flexibility, how is it able to facilitate the transition to a healthy state? Apart from the pursuit of neurological explanations, it is equally important to explore the subjective and common experiences reported when psilocybin is administered. Why do individuals perceive a psilocybin experience as one of the most meaningful in their lives, and why do some describe the experience as “more real” (Griffiths et al., 2006)? Could it be that the meaning found during the experience facilitates normalization instead of the other way around? These questions warrant further investigation in future research endeavours.

With the increasing rise of prevalence of depression, the investigation of its aetiology as well as the exploration of treatment options becomes increasingly important. Through further research, psilocybin-assisted therapy could potentially serve as an alternative for treating depression, particularly for treatment-resistant cases.

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